POTENTIAL IMPACT ASSESSMENT

FitPhone : AI System for Detecting Mental Health Signals from Social Media

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Document History

Version	Date	Status	Author	Description
0.1	12-03-2026	Initial Version	L. Vũ	Initial version
1.0	13-03-2026	Completed First Version, without the reflection.	Complete Group	Reformatted and updated version
2.0	04-06-2026	Revised. Second Version	C. Stijlaart	Revision with TICT Tool, addition transformative reflection.

Introduction

The proposed AI system aims to analyse smartphone usage behaviour, such as screen time, application usage, device pickups, and usage patterns, to identify habits that may affect digital wellbeing. The system uses machine learning to recognise behavioural patterns and provide users with insights and recommendations that encourage healthier smartphone usage. The system would use datasets containing behavioural data (e.g., screen time or usage patterns) and textual content (e.g., posts from platforms such as Reddit or X) to detect patterns associated with decreased mental wellbeing.

While such an AI system could provide valuable insights for research and public health discussions, it also raises important ethical, social, and technical considerations. This impact assessment therefor examines the potential benefits, risks, and broader implications of developing and deploying such a system.

Purpose & Context

Impact on society

The proposed AI system aims to increase awareness of how social media behaviour and smartphone usage may relate to mental wellbeing. By analysing behavioural and text data, the technology could help identify smartphone usage patterns that may contribute to unhealthy digital habits and support users in developing more balanced technology use. In the long term, insights from such systems could support discussions about healthier technology use and improved mental health awareness.

However, the technology also raises important concerns. Analysing personal behaviour or online communication may create risks related to privacy, misinterpretation of data, or unfair profiling of individuals. Therefore, it is important that the system is used carefully and transparently, and that it is positioned as a tool for research and awareness rather than as a system that diagnoses mental health conditions.

Domain Understanding

This project is related to the domains of digital wellbeing and mental health research, where researchers study how smartphone and social media behaviour may relate to mental health. Traditionally, mental health issues are identified through clinical assessments, surveys, or consultations with professionals. However, researchers increasingly explore whether patterns in social media posts or usage behaviour can provide early signals of emotional distress.

Existing research often uses techniques such as natural language processing (NLP) to analyse text data from online platforms. While this can provide useful insights, interpreting online behaviour is challenging because posts may not accurately reflect a person's real mental state and can be influenced by context, humour, or cultural differences.

Because of these limitations, the system should be considered a research or awareness tool rather than a diagnostic system, and careful attention must be given to privacy and responsible data use.

Stakeholders

The proposed AI system may affect multiple groups both directly and indirectly. Identifying these stakeholders helps to better understand the potential societal implications and ensures that different perspectives are considered during development.

Primary Stakeholders

The primary stakeholder is Fitphone, represented by H.J. Kisjes, who initiates the project and makes key decisions about the system's design, data use, and purpose. Fitphone is responsible for ensuring the system is developed ethically and respects privacy and responsible AI principles.

Secondary Stakeholders

Secondary stakeholders are users who want to better understand their smartphone usage habits and improve their digital wellbeing. They may benefit from insights into their digital habits but could also have concerns about privacy or being monitored.

Indirect Stakeholders

Indirect stakeholders include researchers, mental health institutions, technology companies, and policymakers. Researchers may use the insights to understand patterns between digital behaviour and mental health, while institutions could explore preventative wellbeing strategies.

Impact on Stakeholders

Users

- May benefit from increased awareness of how social media behaviour and screen usage relate to their mental wellbeing.
- Could become more conscious of unhealthy digital habits and adjust their behaviour accordingly.
- May experience concerns about privacy, data collection, or being evaluated based on their online behaviour.

Researchers

- Gain access to potential insights into behavioural patterns related to smartphone usage and digital wellbeing.
- Must ensure that datasets are used responsibly and that ethical guidelines for research are followed.

Technology companies

- Could use research insights to design platforms that support healthier usage patterns.
- May face pressure to adjust engagement-focused algorithms that encourage excessive use.

Policymakers

- May use findings to support evidence-based discussions or regulation related to digital wellbeing and social media platforms.

Reflection and Possible Improvements

After mapping the stakeholder landscape, it becomes clear that the system affects more groups than just the direct users. Therefore, it would be important to ensure transparent communication about how the system works, how data is collected, and how insights are generated.

Societal Impact

This section evaluates how the AI system may affect individuals and society, highlighting both positive and negative impacts across privacy, data, transparency, inclusivity, sustainability, misuse risks, and human values.

Privacy

Positive impacts

- If designed responsibly, the system can demonstrate how AI projects can comply with data protection regulations such as the Digital Services Act and general privacy principles.
- Using publicly available and anonymized datasets reduces risks related to personal data misuse.

Negative impacts

- Users may feel uncomfortable or monitored if they believe their online behaviour is being analysed.
- Analysing behavioural usage data may reveal personal habits and routines, such as sleep schedules and device usage patterns., even if anonymised.
- Misuse of personal data could lead to erosion of trust in digital platforms.
- Even publicly available social media data can contain sensitive information about individuals.
- Improper data handling could violate privacy laws or ethical research standards.
- If the system were used in real-world applications, strict safeguards would be necessary to protect user identity and consent.

Data

Positive impacts

- Regular evaluation and validation of datasets is necessary to maintain reliability and fairness.
- The system could demonstrate how machine learning and natural language processing can be applied to analyse social media behaviour.
- It may help researchers identify patterns in large datasets that would be difficult to detect manually.
- The project can contribute to improving AI methods for sentiment analysis and behavioural pattern detection.
- Biases can be reduced through careful curation of datasets and monitoring.

Negative impacts

- Incorrect predictions may lead to inaccurate recommendations or misinterpretation of user behaviour.
- The system may oversimplify complex psychological conditions, which are influenced by many factors beyond social media.
- Biases in datasets may lead to skewed or unrepresentative insights.
- AI models may produce inaccurate predictions due to limited or biased training data.
- Language models may misunderstand sarcasm, slang, or cultural context in social media posts.
- Results should be interpreted as behavioural insights rather than objective assessments of wellbeing.

Transparency

Positive impacts

- Clear communication about the system’s purpose, capabilities, and limitations is essential.
- Stakeholders should know that the system identifies patterns, not diagnoses.
- Ethical design principles such as transparency and data protection can be integrated from the beginning.

Negative impacts

- Users may not understand how their data is analysed or how conclusions are drawn.
- Predicting mental health conditions from online behaviour raises ethical concerns about autonomy and consent.

Inclusivity

Positive impacts

- The AI system could encourage discussions about responsible technology design and healthier social media ecosystems.

Negative impacts

- There is a risk of stigmatization if individuals or groups are categorized as “high risk” based on their online activity.
- Underrepresented groups may not be accurately reflected in the data, leading to biased outcomes.
- The system should be designed to avoid unfair exclusion based on age, culture, or digital literacy.
- The system might unintentionally reinforce biases if the datasets are not diverse or representative.

Sustainability

Positive impacts

- Efficient algorithms and responsible resource usage can minimise environmental impact.
- Sustainability planning supports long-term viability without unnecessary environmental burden.

Hateful and Criminal Actors

Positive impacts

- Designing the system with strong ethical guidelines and safeguards can demonstrate responsible AI practices.
- Raising awareness about potential misuse can help researchers, policymakers, and developers anticipate risks and implement protections.
- The system can serve as a case study for designing AI that minimises harm from malicious actors.

Negative impacts

- The system could be misused to monitor or profile individuals without consent.
- Insights could potentially be exploited to target vulnerable groups or manipulate behaviour online.
- Safeguards and ethical guidelines are necessary to prevent harmful use.

Human Values

Positive impacts

- The AI system could help researchers better understand the relationship between social media behaviour and mental health.
- It may support healthier smartphone habits by identifying usage patterns and encouraging reflection on digital behaviour.
- The results could support public awareness about the effects of excessive or negative social media exposure.
- Insights generated by the system could contribute to healthier digital environments and inform organisations working on digital wellbeing.
- Users' autonomy should be respected, ensuring they are not coerced or misled by the system.
- The project may contribute to responsible AI research by exploring the ethical implications of predicting mental health indicators.

Future Scenarios

Utopian Future

In 2055, society has become more aware of the impact technology has on attention, wellbeing, and daily life. Digital wellbeing systems such as FitPhone help people develop healthier relationships with technology by encouraging reflection rather than addiction. Technology companies increasingly prioritise user wellbeing alongside business goals, resulting in healthier digital environments. People have greater control over their attention, and digital wellbeing is considered an important aspect of public health.

Dystopian Future

In 2055, behavioural monitoring has become a standard feature of digital life. Systems originally designed to improve wellbeing are now widely used to predict and influence human behaviour. Governments, employers, and technology companies collect large amounts of behavioural data to optimise productivity, engagement, and consumption. Privacy has significantly decreased, and individuals have less control over how their behavioural data is used. The boundary between supporting users and manipulating them has largely disappeared.

Impact Reflection

Transformative Perspectives

This project improved our understanding of how machine learning can be used to identify patterns in behavioural data and generate meaningful insights. We learned that the quality and representativeness of data have a major impact on model performance and reliability. The project also highlighted that AI systems are not objective by default and can be affected by limitations in the data, model design, and interpretation of results. As a result, we gained a greater appreciation for responsible AI development, including transparency, fairness, and careful validation of model outputs.

In Practice

Throughout the project, we explored how machine learning models can analyse smartphone usage patterns and identify behavioural trends. During development, we discovered that obtaining suitable data and defining meaningful prediction targets were more challenging than initially expected. This required us to refine the project scope and continuously evaluate whether the model outputs were useful and reliable. The experience demonstrated that successful AI development involves not only building models but also understanding data quality, feature selection, model limitations, and the ethical implications of AI-generated insights.

Conclusion

The AI system has the potential to contribute positively to research on digital wellbeing and social media behaviour by identifying patterns that may indicate mental health risks. However, significant ethical, privacy, and technical challenges must be carefully addressed. The system should therefore be used primarily as a research and analytical tool rather than a diagnostic system. Ensuring transparency, data protection, and responsible use will be essential for minimizing potential harm and maximizing the societal benefits of this technology.